DETAILED DESCRIPTION OF THE INVENTION

Clinical and Consumer Implementation Modes

In various embodiments, the penile anatomical geometry profiling system may be configured for deployment in either a clinical implementation mode, a consumer implementation mode, or a hybrid configuration incorporating aspects of both. While the core sensing architecture, reconstruction engine, and geometry profile generation processes remain consistent, software features, data outputs, and integration pathways may vary depending on the intended use environment.

In a clinical implementation mode, the system may generate extended structural reports including high-resolution cross-sectional distributions, curvature analysis vectors, volumetric calculations, and normalized statistical comparisons relative to curated reference datasets. Clinical mode may incorporate enhanced calibration routines, expanded sampling density, and integration with electronic health record systems or secure institutional data repositories. Output formats may include structured diagnostic datasets, clinician-readable summaries, and longitudinal change tracking modules.

In a consumer implementation mode, the system may generate simplified structural summaries, standardized geometry classifications, or encoded structural signatures suitable for device calibration, personalization, or compatibility modeling. Consumer mode may emphasize privacy-preserving representations that abstract explicit dimensional data while retaining derived proportionality metrics. Visualization outputs may include non-explicit graphical representations of structural parameters or standardized structural indices.

In certain embodiments, the device firmware may include configurable operational tiers that selectively enable or restrict reconstruction resolution, comparative analytics, or data export features. Role-based access controls may differentiate clinician-level access from end-user-level access. Data transmission pathways may vary between local encrypted storage, paired companion applications, or institutionally managed servers depending on deployment configuration.

Hybrid implementations may allow the same hardware platform to operate in consumer mode by default while permitting clinical activation through authenticated credential provisioning. Such flexibility may enable scalable commercialization while preserving the technical capacity for regulated or research-based use cases.

The clinical and consumer deployment architectures described herein are not limited to any specific regulatory classification, output format, user interface configuration, or data-sharing protocol, provided that the system maintains the capability of generating a user-specific penile anatomical geometry profile through synchronized time-of-flight distance acquisition and motion-based positional indexing.